

RK500-09 Multi-parameter water quality sensor



RK500-09 multi-parameter water quality sensor adopts an integrated design and can measure up to 8 parameters at the same time. You can choose DO, COD, pH, ORP, EC/salinity. Using RS-485 output, Modbus-RTU communication protocol, data can be directly transmitted to the acquisition platform.

The multi-parameter water quality sensor is equipped with an automatic cleaning device, automatic cleaning interval time and number of automatic cleaning cycles can be set to adapt to the water quality of different degrees of cleanliness. The automatic cleaning device can effectively clean the surface of the sensor, prevent the adhesion of microorganisms, and greatly reduce the maintenance cost. Each sensor is equipped with a quick-plug waterproof connector, which is easy to disassemble and assemble.

The sensor protection cover at the front is used to protect the internal sensor from damage. There are slots around the protection cover, which can effectively prevent large suspended particles and biological damage to the sensor probe, while not affecting the accuracy of measurement.

FEATURES

- On-line & real-time measurement
- High accuracy
- Simple operation and high reliability
- Fast response
- Strong in corrosion resistance

SPECIFICATIONS

Dissolved oxygen sensor	
Measuring range	0~20mg/L
Accuracy	±0.4mg/L
Resolution	0.01mg/L
EC / salinity sensor	
Measuring range	0-5000uS/cm,0~70PSU
Accuracy	±1.5% F.S.
Resolution	0.1 mS/cm ,0.1PSU
pH sensor	
Measuring range	0~14pH
Accuracy	±0.1pH
Resolution	0.01pH
ORP sensor	
Measuring range	-1500mV~+1500mV
Accuracy	±6mV
Resolution	1mV
Temperature	
Measuring range	0~50℃
Accuracy	±0.5℃
Resolution	0.1℃
NH4+	
Measuring range	0~100mg/L
Accuracy	±10%/±2 mg/L
Resolution	0.1 mg/L
Other Information of Multi-parameter sensor	
Output	RS-485(Modbus/RTU)
Cleaning method	Automatic cleaning 5W@12V 12VDC±5%
Power consumption	5W@12V
power supply	5VDC±5%
Cable length	5m(default). other lengths can be customized

This parameter table only provides partial parameter information

ELECTRICAL CONNECTIONS

Connector (cable)	RS485
Red	V+
Black	V-
Blue	RS485A
White	RS485B

Note: This product has been tested and complies with European CE requirements for EMC directive.

OUTPUT CHARACTERISTICS

- **RS485**

If the transmission distance is over 100m, please add a 120Ω terminal matching resistances on the front end and back end of bus interface respectively. Attach communication protocol.

WARRANTY

This product is warranted to be free of defects in materials and construction for a period of 12 months from date of lead time.

Liability is limited to repair or replacement of defective item.

Communication Protocol (MODBUS)

Transmission mode: MODBUS-RTU, Baud rate: 9600bps, Data bits: 8, Stop bit: 1, Check bit: no

Slave address: the factory default is 06H (set according to the need, 01H to 7FH)

- **The 03H Function Code Example: Read The Value**

Host Scan Order (slave address: 0x06)

06 03 00 00 00 0E C5 B9

Slave Response

06 03 1C 01 04 00 01 00 00 00 00 00 00 00 00 00 6D 00 00 03 71

00 02 00 00 00 00 00 90 00 02 9B 0C

Temp: (0104)H = (260)D, $260/10 = 26(^{\circ}\text{C})$

EC: (006D)H = (109)D, 109(uS/cm)

PH: (0371)H = (881)D, $881/100 = 8.81$

DO: (0090)H = (144)D, $144/100 = 1.44(\text{mg/L})$

- **The 06H Function Code Example: Modify the slave address**

Host Scan Order (Changed to 01H, read and write address must be 06H)

06 06 2002 00 01 E38D

Slave Response:

06 06 2002 00 01 E38D

The slave address is stored in the form of an integer

